Rewriting Eq. 30-10 using Euler's relation results in the most common way of expressing a complex number in polar notation, a **complex exponential**:

EQUATION 30-12
Exponential form of complex numbers. The rectangular form, on the left, is equal to the exponential polar form, on the right.

$$a + jb = M e^{j\theta}$$

Complex numbers in this exponential form are the backbone of DSP mathematics. Start your understanding by memorizing Eqs. 30-8 through 30-12. A strong advantage of using this exponential polar form is that it is very simple to multiply and divide complex numbers:

EQUATION 30-13
Multiplication of complex numbers.

$$M_1 e^{j\theta_1} M_2 e^{j\theta_2} = M_1 M_2 e^{j(\theta_1 + \theta_2)}$$

EQUATION 30-14
Division of complex numbers.

$$\frac{M_1 e^{j\theta_1}}{M_2 e^{j\theta_2}} = \left[\frac{M_1}{M_2}\right] e^{j(\theta_1 - \theta_2)}$$

That is, complex numbers in polar form are multiplied by multiplying their magnitudes and adding their angles. The easiest way to perform addition and subtraction in polar form is to convert the numbers to rectangular form, perform the operation, and reconvert back into polar. Complex numbers are usually expressed in rectangular form in computer routines, but in polar form when writing and manipulating equations. Just as *Re*( ) and *Im*( ) are used to extract the rectangular components from a complex number, the operators *Mag*( ) and *Phase*( ) are used to extract the polar components. For example, if $A = 5e^{j\pi/7}$, then *Mag*(*A*) = 5 and *Phase*(*A*) = $\pi/7$.

Using Complex Numbers by Substitution

Let's summarize where we are at. Solutions to common algebraic equations often contain the square-root of a negative number. These are called *complex numbers*, and represent solutions that cannot exist in the world as we know it. Complex numbers are expressed in one of two forms: $a + bj$ (rectangular), or $Me^{j\theta}$ (polar), where *j* is a symbol representing $\sqrt{-1}$. Using either notation, a single complex number contains two separate pieces of information, either *a* & *b*, or *M* & θ. In spite of their elusive nature, complex numbers follow mathematical laws that are similar (or identical) to those governing ordinary numbers.

This describes what complex numbers are and how they fit into the world of pure mathematics. Our next task is to describe ways they are useful in science